Okan Berhoğlu

Robotics Engineer

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Profile

Robotics Engineer with hands-on experience in robotics research, industrial software development, and autonomous systems. Skilled in software development, industrial communication, data acquisition, simulation, and system integration, with proven experience building and deploying automation and digitalization solutions. Mechanical Engineering background with an M.S. in Robotics and a self-taught software engineering foundation.

Education

Middle East Technical University
Ankara, Turkey
Master of Science, Robotics
2024 – 2026

TOBB Economics and Technology University
Ankara, Turkey
Bachelor of Science, Mechanical Engineering
2018 – 2023

Atatürk University
Remote
Associate Degree as Second Degree, Computer Programming
2021 – 2023

Skills

Programming Languages: Java, C#, Python, C++, JavaScript
Robotics & Simulation: ROS/ROS2, Gazebo, PyTorch MATLAB, Simulink, Simscape, Unity, System Identification
Industrial & Data Systems: Industrial Machine Communication, SQL, InfluxDB
Tools & Platforms: Git, Linux, CUDA, SolidWorks, Docker
Languages: Turkish – Native, English – Upper-Intermediate: TOEFL iBT 2026 score: 5/6 (equivalent to 100/120 in old score system)

Work Experience

Middle East Technical University *Ankara, Turkey*

Graduate Student Researcher *October 2025 – Present*

- Researching autonomous navigation for mobile robots by developing system models and implementing motion planning algorithms and model predictive control techniques to enable real-time obstacle-aware navigation.
- Investigating human rhythmic behavior through a haptic experimental setup, collecting motion data during predefined tasks, and applying system identification to characterize underlying motor control mechanisms.

Teknopar Industrial Automation *Ankara, Turkey*

Software Engineer *February 2024 – October 2025, 1 year 9 months*

- Built and maintained a Windows service-based data acquisition application in C# to enable real-time factory floor visibility across a wide range of factory machines, including CNC controllers, robots, and other industrial equipment, by implementing TCP/UDP-based industrial communication; ensured reliable, continuous operation through version tracking, structured logging, and automated error recovery to support continuous operation.
- Developed an OSGi application using the Java programming language, with a Swing user interface, for military truck systems, enabling operators to monitor and control onboard power units and the vehicle's electrical system in real time, by integrating industrial communication protocols including EtherNet/IP and Modbus TCP for device communication and data collection.
- Secured and delivered a TÜBİTAK funded project by co-authoring the technical proposal covering system modeling, autonomy stack, and digital twin design for a mobile robot and led the technical implementation of AI-based environmental perception and autonomous navigation using ROS2, PyTorch, 3D LiDAR, and depth camera sensors.

Candidate Engineer *December 2022 – February 2024, 1 year 3 months*

- Supported R&D efforts by collecting and analyzing data from CNC machines and industrial systems to inform data acquisition design.
- Contributed to modeling an electromechanical system in Simulink and Simscape for a defense industry project, as part of a team integrating it with IBM Rhapsody so systems engineers could assess design changes against requirements without manual recalculation. Also contributed to validating the model against real-world hardware data through grey-box system identification to make the model more realistic.

Intern *May 2022 – December 2022, 8 months*

- Assisted early-stage industrial communication studies by researching data acquisition methods for CNC machines and industrial equipment.
- Simulated and analyzed real-world motion of an electromechanical system for a defense industry project by developing a MATLAB Simscape Multibody model that integrated CAD geometry with dynamic parameters.
- Created 3D models of hydraulic system components in SolidWorks, integrating them into the full system assembly to support customer presentations and downstream design of related parts.

Anova R&D Technologies *Ankara, Turkey*

Intern *May 2021 – August 2021, 4 months*

- Gained hands-on experience in machining, quality control, and finishing processes, and contributed to mold and fixture design activities.